Curriculum and Instruction

UNIVERSITY OF MINNESOTA
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M. Ed. / Initial Licensure in Science Education: Grades 9-12 Earth & Space Science
Prerequisite Course Evaluation

Candidates are required to have an undergraduate degree from an accredited institution with a major in earth science or closely related field, **or** meet the prerequisites below, and with appropriate liberal arts/general education coursework. Courses in each of the content areas are required to meet Standard 1 of MNSEPT: "*The learner understands and can create learning experiences that make these aspects of content matter meaningful for students" (https://www.revisor.mn.gov/rules/?id=8710.2000).* Any in-progress coursework would be counted as complete if an official transcript with proof of enrollment is provided at the time of application.

Candidate Name:	**UMN ID#:**	**Email:**	**Date:**
Degree Granting Institution:	**Major(s):**	**Minor(s):**	**Prepared by:**

Earth and Space Science Major Status	**Major**	**Degree Granting Institution/ Degree Year**	**Earth& Space Degree Requirement Check ✔**
• An undergraduate Earth & Space science science degree from an accredited institution in the U.S. or an equivalent 4-year Earth & Space science undergraduate degree from an accredited international institution.			

CANDIDATES WITHOUT AN EARTH SCIENCE MAJOR MUST MEET COURSES FOR AN EQUIVALENT OF AN UNDERGRADUATE MAJOR. Per U of MN program requirements, 39 credits are required for an Earth Science degree (including General Psychology). Adjustment can be made for Non U of M degree programs with varying credit requirements.

Content Area	**Requirement**	**U of M Course Options**	**Course/Term/Year**	**Credits/Grade**	**Institution If Other Than Degree Granting**
Psychology	General Psychology (4 credits)	PSY 1001			
Astronomy (with lab)	Astronomy (4 credits)	AST 1001			
Physical Geology (with lab)	Physical Geology (4 credits)	ESCI 1001 OR ESCI 1012 OR ESCI 2201 OR ESCI 2301*			
Historical Geology	Earth History, or	ESCI 2202 OR			

	Geochronology & Earth History (3 Credits)	ESCI 4102W OR ESCI 4103W			
Oceanography or Limnology	Oceanography, or Limnology (4 credits)	ESCI 1006 OR ESCI 1106 OR ESCI 5705			
Earth Materials	Mineralogy, or Petrology (3 credits)	ESCI 2301* ESCI 2302			
Land Forms	Geomorphology, or Glacial Geology (4 credits)	ESCI 3402 OR ESCI 4701 OR ESCI 4703			
Human-Oriented Geology (two courses) (6 Credits)	Climate Change and Human History	ESCI 3002 OR ESCI 3402			
	Water and Society	ESCI 3004 OR ESCI 4702 OR ESCI 4801*			
Field Methods of Geology (one course) 4 Credits	Introduction to Field Geology (4 credits)	ESCI 3911			
	Advanced Field Geology (4 credits)	ESCI 4911			
	Field Hydrogeology (4 credits)	ESCI 4971W			
Integrated Earth Systems (3 credits)	Earth Systems: Geosphere/Biosphere Interactions (3 credits)	ESCI 2202 OR ESCI 2203 OR ESCI 3303W OR ESCI 4401 OR ESCI 4801*			
Research in Earth Science (3 credits)	Problems in Geology & Geophysics, *or* Senior Thesis	ESCI 3896 OR ESCI 4093 OR ESCI 4094			

*ESCI 2301 and 4801 may each only be used once toward prerequisites.

Additional Licensure in Grades 5-8 General Science (Optional)
Prerequisite Course Evaluation

Content Area	Requirement	U of M Course Options	Course/Term/Year	Credits/Grade	Institution
Life Science two courses (with labs)	Biology I	BIOL 1001 or 1009 or 1951/1961			
	Biology II	BIOL 2003/3004 or 1055 or GCD 3022 or 3033			
Physics sequence two courses (with labs)	Physics I	PHYS 1101 or 1221 or 1301W			
	Physics II	PHYS 1102 or 1222 or 1302W			
Chemistry sequence	General Chemistry I	CHEM 1061 & 1065 or			

two courses (with labs)		1081 & 1065 or 1071H & 1075H			
	General Chemistry II	CHEM 1062 & 1066 or 1082 & 1086 or 1072H & 1076H			
Earth science two courses (with labs)	Physical Geology	ESCI 1001			
	Astronomy	AST 1001			